

8 Arginase is an enzyme produced in liver cells.

It breaks down arginine into urea and ornithine.

A mutation in a gene called ARD1 can cause arginase deficiency (AD), which can be fatal. The mutant allele is recessive.

(a) It is estimated that 1 in 300 000 to 1 000 000 individuals have AD.

(i) It was suggested that this mutant allele arose in meiosis during prophase I.

Suggest how this mutant allele could be produced in prophase I.

(1)

(ii) If the incidence of AD was 1 in 300 000, and the population was 7 800 000 000, then 26 000 people would have AD.

Calculate how many people do not have a mutant allele for ARD1.

Use the equation:

$$p^2 + 2pq + q^2 = 1$$

(3)

Answer



- (b) One treatment for AD is mRNA therapy. In this therapy, people are given the mRNA molecules needed for arginase production.

The mRNA used in this therapy is active mRNA.

Active mRNA is produced by post-transcriptional changes to mRNA (pre-mRNA).

Give **two** differences between active mRNA and pre-mRNA.

(2)

1

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2

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- (i) Explain how a pluripotent stem cell could become a cell that synthesises arginase.

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- (ii) Explain why society may support the use of hiPSC stem cells to treat patients with AD.

(3)

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(Total for Question 8 = 14 marks)

TOTAL FOR PAPER = 80 MARKS

